Bayesian Statistics One-Shot

Tim Book

GENERAL ASSEMBLY

Agenda

- What is Frequentism? What is Bayesianism?
- What is Bayes' Theorem?
- How can we apply Bayes' Theorem to statistical modeling?
- Decomposing conditional distributions
- Fitting Bayesian models using PyMC!

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You've heard of this battle...

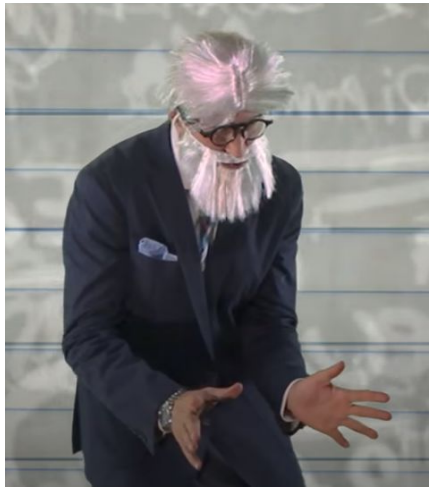




Source: https://www.youtube.com/watch?v=uHGlCi9jOWY

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But how about this one?!



Frequentist



Bayesian

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Statisticians are still infighting over this one...

Up until now, we've only concerned ourselves with mainstream **frequentist** beliefs. A frequentist approaches problems the following way:

1. Begin with some parametric assumptions about nature
2. Collect data
3. Use that data to make inferences about the population

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So what's the problem?

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Shortcomings of Frequentism

Sounds intuitive right? Make some assumptions, gather data, use data! But there are some issues:

- This makes it impossible to study phenomena that we can't collect data on because they only happen once (eg, elections)
- This makes it difficult to study phenomena associated with expensive experiments (or any other reason we might have small samples)
- Susceptible to bad luck - what if we just get really weird results with our small, expensive experiment?
- The only information we can use in our calculations are data - what if we have other types of knowledge that can help?

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Bayesian Statistics: An Alternative Approach!

We can alleviate these shortcomings by using **Bayesian** techniques. A Bayesian might approach a problem the following way:

- Begin with some parametric assumptions about nature
- Assign **a priori** beliefs to parameters that represent our prior knowledge and our degree of strength in that belief
- Collect data
- Combine our priors with our data to get **posterior estimates** of our parameters of interest

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Bayesian Statistics in a Nutshell

Priors + Data = Posterior Estimates

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But first: a prereq

Probability Review + Bayes Theorem



Remind me: What is this probability rule?

Suppose I have two events A and B. How do I compute the probability of both A and B occuring?

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Probability Law: “and”

$$P(A \cap B) = P(A)P(B|A)$$
$$= P(B)P(A|B)$$

(Since “A and B” is the same as “B and A”, these two are equivalent!)

And if we rearrange...

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$= \frac{P(A|B)P(B)}{P(A)}$$

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Note that we flipped the conditional here!

Another quick recap

Probability Mass/Density Functions

pmfs/pdfs

Every random variable has a **probability mass (or density) function** that tells us where its probability is spread out. They're typically denoted with a lowercase *f*. In the case of discrete variables, the pmf is defined like this:

$$P(X = x) = f(x)$$

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Next, we need to talk about parameters

In frequentist methods, population parameters are **fixed constants**. After all, there's only one true population mean for a real life phenomenon.

In Bayesian statistics, we loosen this assumption. **We let population parameters be random variables!** This allows us to assign **prior information** to them in our models.

We denote all of our random variables combined as $\boldsymbol{\theta}$ in the next slides. We'll denote the data as $\boldsymbol{x}$.

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Rewriting Bayes' Theorem with Random Variables

$$f(\theta|\mathbf{x}) = \frac{f(\mathbf{x}|\theta)\,f(\theta)}{f(\mathbf{x})}$$

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Rewriting Bayes' Theorem with Random Variables

priors (we assume this based on prior knowledge)

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Rewriting Bayes' Theorem with Random Variables

Likelihood (not really an assumption, but has to do with how we parametrize the problem)

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Marginal distribution of our data (ignore this)

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Posterior distribution (our goal!)
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Marginal distribution of our data (ignore this)

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We can ignore that denominator if we believe hard enough

$$f(\theta|\mathbf{x}) \propto f(\mathbf{x}|\theta)f(\theta)$$

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$$f(\theta|\mathbf{x}) \propto f(\mathbf{x}|\theta)f(\theta)$$

Proportional to

Must integrate to 1 to be a valid pmf/pdf (calculus alert!)

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Let's talk about calculation here...

While this looks like easy multiplication, this can only be done by hand in a few simple (unrealistic) scenarios (see: **conjugate priors**). In general, to compute the thing on the left here, some form of **Markov Chain Monte Carlo (MCMC)** simulation must be carried out. The top contender for doing this in Python is with the **PyMC** library!

$$f(\theta|\mathbf{x}) \propto f(\mathbf{x}|\theta)f(\theta)$$

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Let's take a step back...

I don't want to lose sight of what we're doing here. Let me try to simplify it with bullet points:

- We want to calculate a **posterior probability distribution** of parameters we care about. This takes into account both our priors and our data.
- It's basically impossible to compute the pmf/pdf of the posterior by hand.
- Instead, **we use fancy sampling techniques to sample from the posterior without ever actually calculating it!**

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FiveThirtyEight: A Motivating Example

FiveThirtyEight's Election Models

Famous Bayesian and Twitter curmudgeon **Nate Silver** talks about how he uses Bayesian techniques to fit his highly accurate election forecast models.

This situation is ripe for Bayesian techniques because the election itself only happens once.



source: slate.com

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FiveThirtyEight's Election Models

He says that his prior assumptions use one key fact that we always forget early in election cycles:

elections usually end up pretty close to 50/50.

He also uses poll data as data that formulate his likelihood. However, he also grades polls based on their quality. **These are more a priori assumptions!**



source: slate.com

My Best Guess at 538's Election Models

We'll let *p* denote the probability of candidate A winning.

$$f(p|\mathbf{x}) \propto f(\mathbf{x}|p,\theta)f(p)f(\theta)f(\text{other stuff})$$

Posterior prediction of probability of candidate A winning

Likelihood (proprietary to 538)

"Elections usually end up close to 50/50"

"We give this poll a C rating, that poll an A rating…"

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Let’s try some of our own out!



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